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Roddy Taing

Applied Scientist | U.S. Citizen

Willing to Relocate

EDUCATION

Ph.D. Statistics | University of Wisconsin-Madison August 2021 — December 2027
Master's (M.S.) Statistics | University of Wisconsin-Madison December 2024
Bachelor's (B.S.) Mathematics, B.S. Physics | University of Connecticut May 2021

EXPERIENCE

Data Science Institute Graduate Student Intern June 2026 — September 2026
Lawrence Livermore National Laboratory Livermore, CA

- Analyzed material strength data in collaboration with material scientists using Bayesian mixed effects models for uncertainty quantification.
- Developed agentic AI workflows to facilitate automated defect detection in advanced manufacturing using CT scan image data.

Research Intern June 2025 — August 2025
STR (defense/national security firm) Woburn, MA

- Designed orbital mechanics simulations to model the viability and real performance of a space-based sensing system (**MATLAB**)
- Analyzed the data and presented the analysis for these simulations, which were used in the continuation of a 60+ million USD project and to secure grant funding for follow-up work

Research Intern June 2023 — September 2023
Raytheon Technologies | BBN Arlington, VA

- Researched optimal sensor tasking and sensor fusion methods for predicting future satellite dynamics
- Designed orbital mechanics simulations to verify research findings. These findings were presented to customers at the AMOS conference to help secure future contracts (**Python**) [1]

Graduate Research Assistant August 2021 — Present
University of Wisconsin-Madison | Department of Statistics Madison, WI

- Develop representation learning methods for large, network-structured datasets.
— Published resulting methodology at ICML. [2]
- Developed image segmentation tools for determining white matter and grey matter regions in the brain based on deep learning methods via **Keras/TensorFlow**
- Mentored undergraduate students and foster growth of their technical skills
- Engage in general teaching duties (teaching assistant)

Data Scientist June 2021 — August 2021
Bio-Techne | Data Lab Minneapolis, MN

- Developed product recommendations for tens of thousands of products using knowledge graph/network embeddings (**PyTorch** in **Databricks, Apache Spark**) Pushed to production and increased click-through-rate by 20%
- Contributed to the development of a nonlinear regression model for time-series forecasting of revenue for use by Head of Finance and Chief Financial Officer (**Python**). Pushed to production
- Developed attribution models to determine effectiveness of marketing campaigns based on various data sources (**Python**). Pushed to production
— Accurately attributed sales orders to marketing campaigns with 94% accuracy
- Utilized natural language processing techniques to predict future scientific research trends and anticipate product demand (**Python/TensorFlow/Keras**)
- Created customer personas via unsupervised machine learning methods to guide targeted marketing strategy (**Python**)

Data Scientist January 2017 — August 2021
University of Connecticut | Institute for Systems Genomics | Department of Physics Storrs, CT

- Established a data analysis pipeline to extract relevant biological findings from terabytes of high-dimensional, unstructured genomic data
— Pipeline included the use of unsupervised and supervised machine learning methods
- Developed and implemented a quality control method using a mixture model for identifying and filtering poor or incomplete scRNA-seq datasets
- Created a variational autoencoder for visualizing data with complex topologies (**Python/Keras/Tensorflow**)
- Modified and implemented various clustering algorithms (K-means, DBSCAN, and hierarchical) to large datasets for cell-type identification (**Python**)
- Designed and implemented a convolutional neural network using **Keras/TensorFlow/OpenCV/scikit-learn** APIs in **Python** for detection of diffraction patterns with low signal in noisy backgrounds

PROGRAMMING LANGUAGES AND OPERATING SYSTEMS

Python, R, C++, Bash, SQL, Mathematica, and MATLAB | macOS, Linux, Windows

TECHNIQUES

Machine Learning (supervised and unsupervised), regression, computer vision, network analysis, natural language processing, statistical analysis, digital signal processing, Fourier analysis, and simulations.

PROGRAMS/PLATFORMS/TECHNOLOGIES

Apache Spark, PySpark, Databricks (distributed computing), BigQuery (SQL), Microsoft Azure, Docker, Bazel, Git, GitHub

PUBLICATIONS

1. Vanslette, K. & Taing, R. *Toward Optimal Conjunction-Based Sensor Tasking Using Inferential Moments* in *Proceedings of the Advanced Maui Optical and Space Surveillance (AMOS) Technologies Conference* (2023).
2. Taing, R. & Levin, K. *On the Effect of Misspecifying the Embedding Dimension in Low-rank Network Models* in *Forty-third International Conference on Machine Learning* (2026).